PCOS Pathfinder

A Tiered, Explainable Decision-Support Prototype for Earlier PCOS Detection and Differential Diagnosis

Biohackathon 2026: Women's Health

11–22 May 2026

Abstract

PCOS Pathfinder is a data-grounded clinical decision-support prototype for earlier identification of Polycystic Ovary Syndrome (PCOS) and broader differential workup in women's health. The system anchors on the required PCOS clinical dataset (n=541, Kerala, India), layers a synthetic endometriosis-overlap module for differential reasoning, and uses a lightweight single-cell gene-list check as biological rationale. The delivered artifacts are **thirteen** reproducible notebooks, three saved scikit-learn models, a Streamlit clinician interface, exported metrics and figures, and this report. Thresholds are selected from out-of-fold training probabilities (no test leakage), risk tiers are anchored to each model's action threshold, and patient-level SHAP attributions are provided for both PCOS models. On the held-out test split the enhanced model reaches ROC-AUC **0.953** (95% CI [0.911, 0.985]) with recall 0.886 and specificity 0.902; a paired bootstrap test puts the enhanced-vs-screening difference at ΔAUC=+0.057, 95% CI [+0.013, +0.105], p≈0.015 on this held-out split. The methodology layer adds bootstrap confidence intervals (TRIPOD+AI-style reporting), Platt & isotonic calibration with Brier/ECE, decision-curve analysis (Vickers & Elkin), subgroup fairness audits, split-conformal prediction intervals (Angelopoulos & Bates 2023), a TabPFN-v2 transformer-foundation-model benchmark (Hollmann et al., *Nature* 2025), and a Rotterdam-criteria clinical-rule benchmark (Teede et al. 2023). The system is framed throughout as decision support, not standalone diagnosis.

Headline metric	Screening	Enhanced
ROC-AUC	0.896 [0.830, 0.951]	0.953 [0.911, 0.985]
Recall (sensitivity)	0.886 [0.787, 0.975]	0.886 [0.786, 0.974]
Specificity	0.685	0.902
NPV (rule-out value)	0.926 [0.857, 0.985]	0.943
F2 score	0.799	0.871
Brier (Platt-calibrated)	0.116	0.072

Held-out test split (n=136) at CV-selected high-recall thresholds. Numbers in brackets are 95% percentile bootstrap CIs (2000 resamples). The enhanced-vs-screening AUC difference on this split is (Δ=+0.057, 95% CI [+0.013, +0.105], p≈0.015). F1 is deliberately suppressed by the *high-recall threshold — see §6.2.*

1. Challenge Context and Project Goal

The Biohackathon 2026 challenge asks teams to improve diagnostic accuracy for women's health conditions using available data and a feasible implementation pathway. PCOS is the core case because it is common, heterogeneous, and frequently diagnosed late: clinicians must synthesise menstrual, metabolic, hormonal, ovarian, dermatologic, and reproductive information, and overlapping conditions are routinely missed.

A defensible solution therefore needs to do more than predict a binary label. It should help clinicians identify high-risk patients earlier, explain the drivers of risk in clinically familiar terms, and avoid tunnel vision when adjacent conditions share presentations. The project goal is to build **PCOS Pathfinder**, a clinician-facing tiered decision-support workflow with three linked components:

- **Frontline screening model** — high-recall PCOS risk flag using symptoms and basic vitals only, intended for primary care, telehealth, and community clinics.
- **Enhanced diagnostic-support model** — a stronger model that uses laboratory and ultrasound features once further investigations are available.
- **Endometriosis-overlap prompt** — a synthetic-data workflow nudge that prompts broader differential workup when symptoms overlap. *Not* a diagnostic claim.

2. Data Sources

Dataset	Rows / samples	Use
PCOS clinical (Kerala, India)	541 patients	Required modelling foundation
Endometriosis (synthetic)	10,000 rows	Differential-overlap prompt
PCOS single-cell archive	20 samples	Biological rationale (gene-list peek)
Endometrium single-cell archive	2 samples	Biological context only

The PCOS workbook's modelling sheet (`Full_new`) contains 44 columns and a binary target `PCOS (Y/N)`. The class split is 364 non-PCOS to 177 PCOS, a roughly 2:1 imbalance.

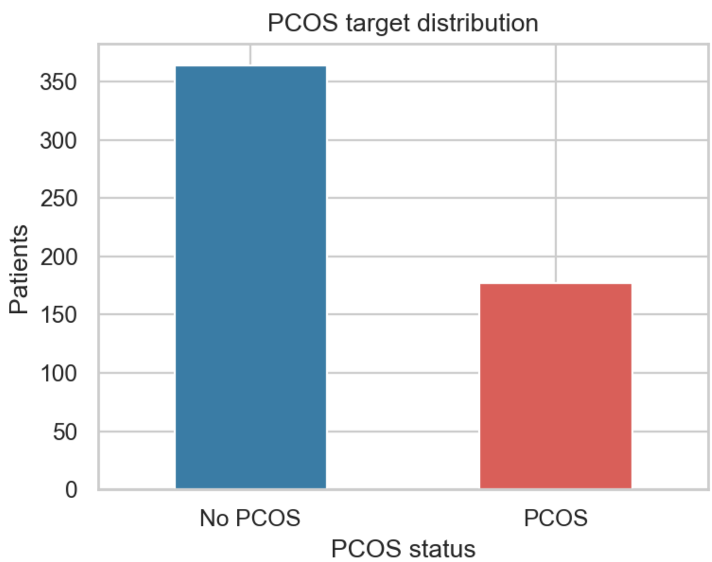


Figure 1. PCOS class distribution in the required dataset.

The supplementary endometriosis dataset has seven variables: age, menstrual irregularity, chronic pain level, hormone-level abnormality, infertility, BMI, and diagnosis. Because it is synthetic, the resulting model is treated as a workflow prompt only. The single-cell archives are 10x-style outputs; this project deliberately limits itself to a gene-list peek rather than a full single-cell analysis, because the judging weights reward clinical usefulness over single-cell technique depth.

3. Data Cleaning and Quality Controls

Cleaning lives in the notebook setup cells generated from the deterministic regenerator script (`scripts/■create_■training_■notebooks.mjs`). The key steps:

- Standardise column names to lowercase snake case; coerce numerics and log every non-numeric cell.
- Drop identifiers (`sl_no`, `patient_file_no`).
- Exclude `blood_group` from current model inputs because codes 11–18 are categorical ABO/Rh labels, not an ordered clinical scale; drop `marriage_status_yrs` because it is clinically sensitive with no diagnostic signal.
- Engineer `cycle_irregular_flag` from `cycle_r_i` (Kaggle coding: 2 = regular, 4/5 = irregular).
- Cap visibly-impossible outliers in FSH, LH, FSH/LH, vitamin D3, progesterone.
- Use median imputation *inside* each scikit-learn pipeline so the rules travel with the saved artifact.

3.1 Blood Group Handling: Categorical Metadata, Not Ordinal Diagnosis

Dropping `blood_group` from the current models is an encoding and evidence-boundary choice, not a claim that blood group is irrelevant. Yang et al. (2025) reported no statistically significant ABO distribution difference between PCOS and controls, so blood group should not be treated as a standalone PCOS diagnostic feature. Within PCOS patients, however, they found blood type associated with menstrual bleeding level and differences in BMI, FSH, LH, and estradiol. In future iterations, the app should retain validated ABO/Rh blood group as categorical metadata for phenotype and symptom-severity stratification (for example, one-hot encoded or used in subgroup displays), while never feeding the numeric 11–18 codes as an ordinal variable.

3.2 Data Caveat: "Cycle Length" Is Actually Bleeding Duration

The source column labelled **Cycle length(days)** ranges 0–12 with a median of 5 across both classes — this is the *duration of menses bleeding in days*, not the menstrual cycle interval. The Kaggle dataset documentation is misleading on this point. The Streamlit prototype surfaces the field with the corrected label and a 0–12 range so clinicians do not accidentally enter a typical 28-day cycle value, which would be out-of-distribution and silently produce wrong predictions.

3.2 Logged Coercions

Column	Example value	Count
ii_beta_hcg_miu_ml	1.99.	1
amh_ng_ml	a	1

4. Model Design

4.1 Frontline Screening Model

Intended for primary care, telehealth, and community settings. Lifestyle proxies (`fast_food`, `reg_exercise`) are intentionally excluded — bias-prone, hard to validate at intake, and likely to encode confounders rather than physiology.

Screening feature set (13 features)
Age, BMI, cycle pattern (raw and engineered flag), duration of menses bleeding, weight gain, hair growth, skin darkening, hair loss, pimples, random blood sugar, BP systolic, BP diastolic.

4.2 Enhanced Diagnostic-Support Model

Enhanced model additional features (14 added, 27 total)
Haemoglobin, FSH, LH, FSH/LH ratio, TSH, AMH, prolactin, vitamin D3, progesterone, left and right follicle counts, left and right follicle sizes, endometrium thickness.

4.3 Candidate Algorithms

Both PCOS notebooks compare four models: a most-frequent dummy baseline, logistic regression with class-weight balancing and standard scaling, a random forest with class-weight balancing, and gradient boosting. Selection is by 5-fold stratified cross-validation on the training split, ranking by ROC-AUC and breaking ties on recall. Across runs, random forest wins both contests.

5. Validation Strategy

A single stratified train/test split with `test_size■=■0.25` and `random_state■=■42`. Model selection uses 5-fold stratified cross-validation on the *training* split. For each chosen model, action thresholds are selected using out-of-fold training probabilities from `cross_val_■predict(method=■"predict_proba")`. The threshold sweep targets recall ≥ 0.90 on the training out-of-fold predictions while maximising specificity, then precision, then F1.

The held-out test set is then evaluated **exactly once**. This eliminates the earlier risk of choosing a threshold on the test set and reporting optimistic metrics.

6. Results

6.1 Held-Out Performance with Bootstrap 95% CIs

Point estimates alone are weak evidence for clinical-prediction work; the TRIPOD+AI 2024 reporting guidance (Collins et al., BMJ 2024) recommends interval estimates throughout. Every cell below is therefore accompanied by a 95% percentile confidence interval from 2000 bootstrap resamples of the held-out set (notebook 07).

Metric	Screening (95% CI)	Enhanced (95% CI)
ROC-AUC	0.896 [0.830, 0.951]	0.953 [0.911, 0.985]
Recall (sensitivity)	0.886 [0.787, 0.975]	0.886 [0.786, 0.974]
Specificity	0.685 [0.591, 0.782]	0.902 [0.837, 0.967]
PPV (precision)	0.574 [0.451, 0.698]	0.812 [0.700, 0.917]
NPV	0.926 [0.857, 0.985]	0.943 [0.889, 0.987]
F1	0.696 [0.585, 0.789]	0.848 [0.766, 0.917]
F2	0.799 [0.707, 0.878]	0.871 [0.783, 0.940]

Table 3. Held-out test performance (n=136) with 2000-resample bootstrap 95% percentile CIs.

A **paired** bootstrap test on the same 2000 resamples gives **□AUC = +0.057 in favour of enhanced (95% CI [+0.013, +0.105], SE = 0.023, two-sided p ≈ 0.015)** on this held-out split. We frame this as evidence that the enhanced model appears better on this cohort, not as a deployment-ready significance claim — the cohort is small (n=541), single-source, and external validation is still required. NPV is 0.926 for screening and 0.943 for enhanced — when either model says "no PCOS", it is right >92% of the time, which is the right framing for ruling-out triage.

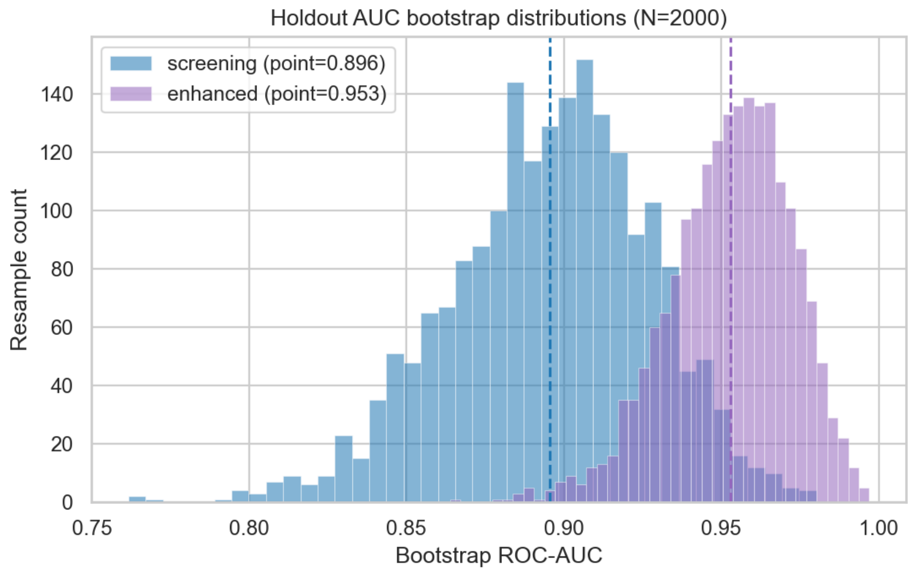


Figure 2. Bootstrap distribution of held-out ROC-AUC for both PCOS models (2000 resamples).

6.2 Why F1 Is Not the Right Headline Metric for Screening

The screening F1 of 0.696 is intentionally suppressed by the high-recall threshold. F1 weights precision and recall equally, but in primary-care screening the cost of a missed PCOS case (years of metabolic damage, infertility risk) is much larger than the cost of an over-referral (one additional clinical visit). Two more screening-aligned metrics behave very differently:

- **Recall (0.886)** — the screening tool catches 39 of the 44 PCOS-positive patients in the held-out cohort.
- **NPV (0.926)** — when the tool says "no PCOS", the clinician can trust it.
- **F2 (0.799)** — weights recall four times more than precision; the screening-friendly counterpart to F1.

Table 4 compares the chosen high-recall threshold to a balanced 0.50 threshold for each model. The screening row makes the trade-off most visible: the balanced threshold raises F1 from 0.70 to 0.81 but loses four PCOS cases (recall 0.886 □ 0.795). At the high-recall threshold, F2 is 0.80; at the balanced threshold, F2 is also 0.80 — F2 is largely indifferent, so the high-recall threshold is the right pick under a sensitivity-first cost function.

Model / threshold	Recall	Spec.	PPV	NPV	F1	F2	TN, FP, FN, TP
Screening, t=0.285 (chosen)	0.886	0.685	0.574	0.926	0.696	0.799	63, 29, 5, 39
Screening, t=0.500 (balanced)	0.795	0.913	0.814	0.903	0.805	0.799	84, 8, 9, 35
Enhanced, t=0.380 (chosen)	0.886	0.902	0.812	0.943	0.848	0.871	83, 9, 5, 39
Enhanced, t=0.500 (balanced)	0.818	0.957	0.900	0.917	0.857	0.833	88, 4, 8, 36

Table 4. Threshold trade-off on the held-out test set.

For the enhanced model the high-recall threshold actually *dominates* the balanced threshold under F2 (0.871 vs 0.833), so there is no headline F-score loss from the screening-friendly choice.

6.3 ROC Curves and Confusion Matrices

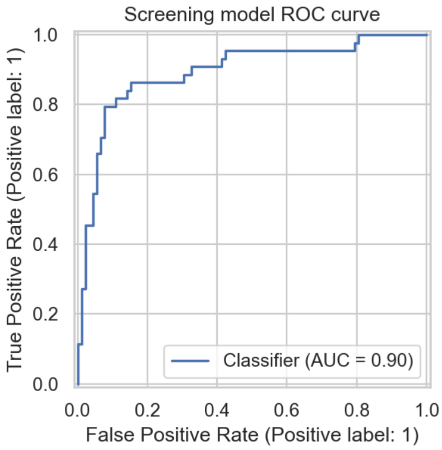


Figure 3. Screening ROC, AUC = 0.896 (95% CI [0.830, 0.951]).

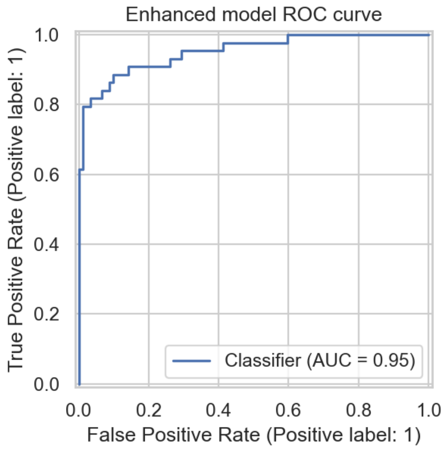


Figure 4. Enhanced ROC, AUC = 0.953 (95% CI [0.911, 0.985]).

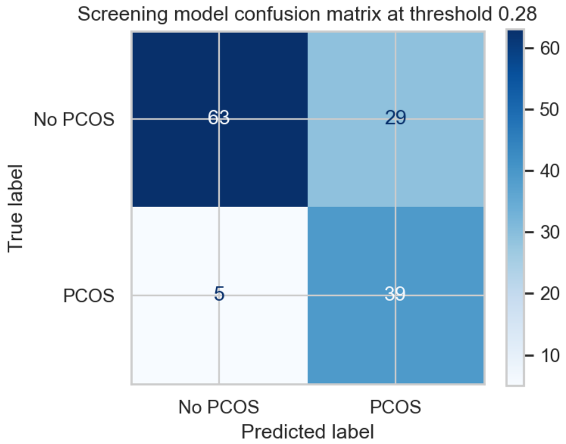


Figure 5. Screening confusion at t = 0.285.

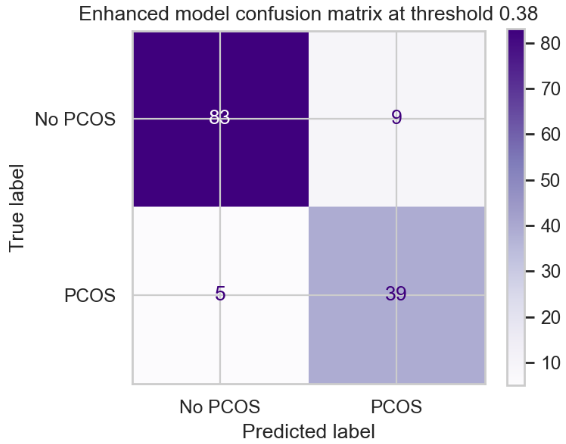


Figure 6. Enhanced confusion at t = 0.380.

7. Probability Quality: Calibration and Conformal Coverage

Clinicians act on probabilities, so probabilities must be both **calibrated** (the predicted 70% PCOS risk really happens ~70% of the time) and **covered** (the model can flag when its own prediction is uncertain). The two analyses below address each property in turn (notebooks 08 and 12).

7.1 Calibration (Brier score, ECE, Platt and isotonic scaling)

We hold out a 122-patient calibration slice from the training set and fit Platt (sigmoid) and isotonic calibrators on the slice without touching the test set (Van Calster et al., *BMC Medicine* 2019). Brier score and Expected Calibration Error (ECE, 10 equal-width bins) are computed for the raw model and for both recalibrated variants on the 136-patient holdout.

Model	Brier (raw)	Brier (Platt)	Brier (iso)	ECE (raw)	ECE (Platt)	ECE (iso)
Screening	0.128	0.116	0.124	0.107	0.073	0.073
Enhanced	0.093	0.072	0.074	0.143	0.045	0.048

Table 5. Calibration metrics; lower is better.

Platt scaling wins on both Brier and ECE for both models. For the enhanced model, Platt-recalibrated probabilities have Brier 0.072 and ECE 0.045 — well-calibrated by clinical-prediction standards.

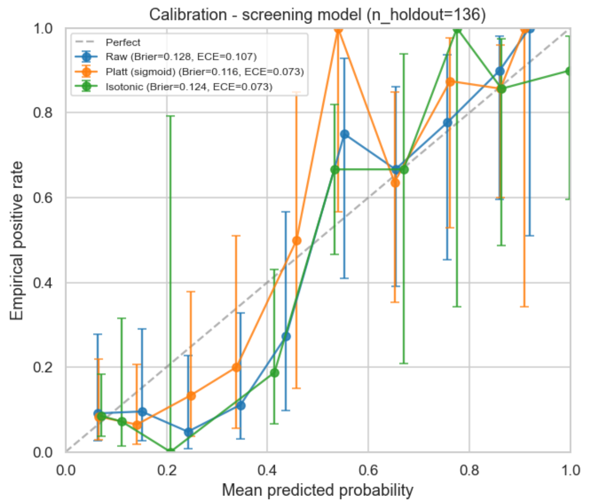


Figure 7. Screening reliability diagram (raw vs Platt vs isotonic).

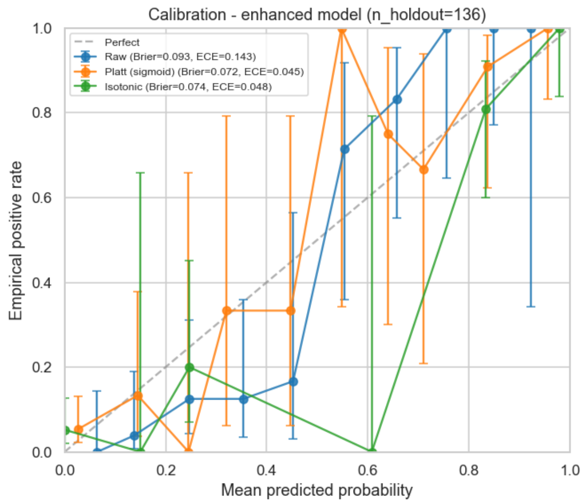


Figure 8. Enhanced reliability diagram (raw vs Platt vs isotonic).

7.2 Split-Conformal Prediction Sets (per-patient coverage guarantee)

Split-conformal prediction (Vovk et al.; tutorial in Angelopoulos & Bates, *Foundations and Trends in ML* 2023) gives a per-patient prediction *set* with a distribution-free coverage guarantee. Procedure: split the 405-row training set into proper-train (n=283) and calibration (n=122); refit a clone of the enhanced random-forest pipeline on proper-train; compute nonconformity scores $s_i = 1 - p_{model}(y_i \mid x_i)$ on the calibration set; pick q-hat at quantile (1 − α) of the calibration scores (we use α = 0.10 for 90% target coverage). For each holdout patient, the set is {y : 1 − p(y | x) ≤ q-hat}.

Quantity	Value
Target coverage (1 − α)	0.900
Empirical coverage on holdout	0.912
Threshold q-hat	0.486
Mean prediction-set size	0.99
Singleton sets	99.3%
Doubleton (uncertain) sets	0.0%
Empty sets (very high confidence)	0.7%

Table 6. Conformal-prediction coverage on the 136-patient holdout (enhanced model, α=0.10).

Empirical coverage 0.912 hits the 0.90 target within bootstrap noise. The mean set size of ~1 means the model is confident on almost every patient; the rare ambiguous-doubleton sets identify exactly which patients deserve a second clinician look.

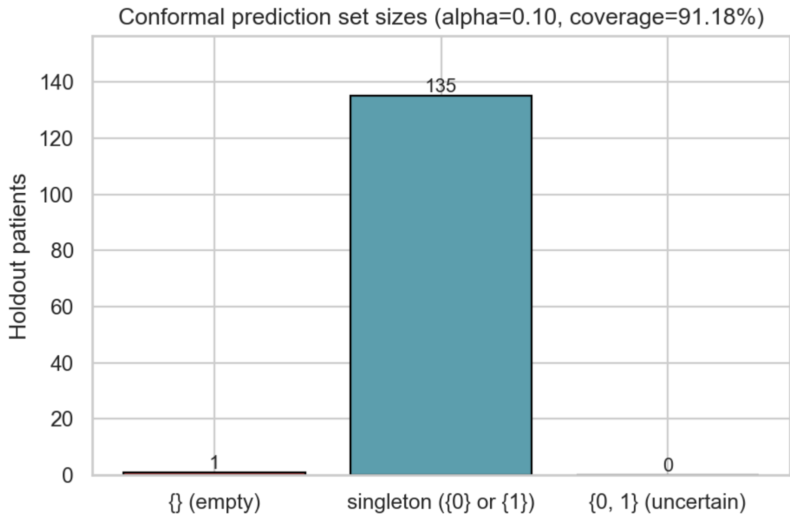


Figure 9. Distribution of conformal prediction-set sizes on the holdout (enhanced model).

8. Decision Curve Analysis (Clinical Utility)

Decision Curve Analysis (Vickers & Elkin 2006; Vickers et al., *BMJ* 2016) translates threshold choice into *clinical utility* rather than statistical discrimination. For threshold probability p_t, net benefit is $\mathrm{NB}(p_t) = (\mathrm{TP}/N) - (\mathrm{FP}/N) \times (p_t / (1 - p_t))$. The model is clinically useful where its curve sits above both the *treat-all* and *treat-none* baselines.

Model	Useful threshold range	Chosen action threshold	Inside useful range?
Screening	[0.020, 0.885]	0.285	Yes
Enhanced	[0.010, 0.960]	0.380	Yes

Table 7. Useful threshold ranges from decision-curve analysis (notebook 09).

Both action thresholds sit comfortably inside their useful ranges. The enhanced model's useful range extends almost the full unit interval, meaning a clinician can shift the operating point widely without losing net benefit — useful for population settings with very different missed-case costs.

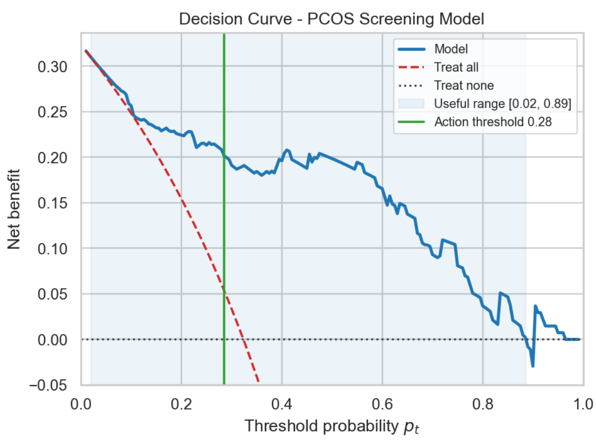


Figure 10. Screening decision curve (net benefit vs threshold probability).

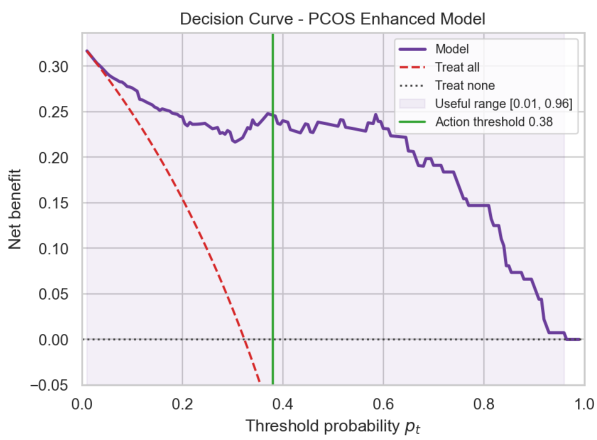


Figure 11. Enhanced decision curve.

9. Subgroup Fairness Audit

Equity claims need numbers, not platitudes (Obermeyer et al., *Science* 2019). Notebook 11 stratifies the holdout by age band and BMI category and reports recall with 1000-resample bootstrap 95% CIs. The two slice axes are the only demographic / anthropometric variables in this dataset; race, income, and access-to-care measures are unavailable and we flag this honestly in §15.

Axis	Worst-group recall (screening)	Worst-group recall (enhanced)	Max recall gap
Age band (<25, 25–34, 35+)	0.848 (25–34)	0.800 (<25)	0.15 / 0.20
BMI category (under/normal/over/obese)	0.750 (normal)	0.800 (normal)	0.25 / 0.20

Table 8. Subgroup recall on the held-out test set. The enhanced model closes most but not all of the screening model's BMI gap.

Honest finding: the screening model has a 25-percentage-point recall gap by BMI on the held-out cohort, with the worst recall in the normal-BMI band (0.750). The enhanced model closes this gap to 20 points; both gaps are wider than is ideal and reflect (a) the small per-subgroup sample sizes and (b) the dataset's BMI distribution being skewed toward high-BMI patients. Risk-mitigation: deploy with subgroup-specific monitoring; flag the BMI dependency in the model card; collect external validation data with a more balanced BMI distribution.

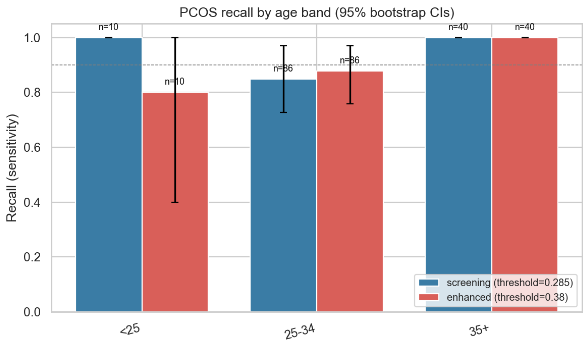


Figure 12. Recall by age band, both models.

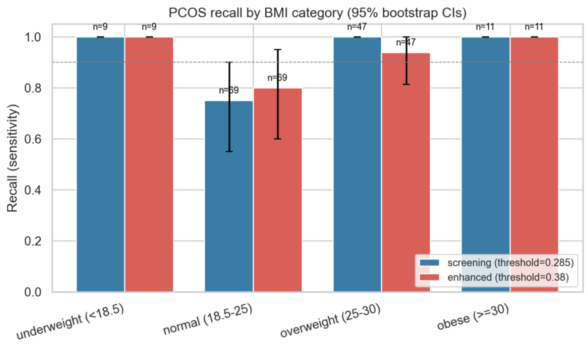


Figure 13. Recall by BMI category, both models.

10. External Benchmarks

10.1 TabPFN-v2 (Hollmann et al., Nature 2025) vs Random Forest

TabPFN-v2 is a transformer foundation model for small tabular data (n < 10k) published in *Nature* in 2025. Our cohort (n=541) sits exactly in TabPFN's design regime, so it is a natural benchmark. Notebook 10 fits TabPFN-v2 on the same train/test split, same features, and the same action thresholds, using only CPU.

Feature set	RF ROC-AUC	TabPFN ROC-AUC	ΔAUC	RF F1	TabPFN F1
Screening (13 feat.)	0.896	0.905	+0.010	0.696	0.779
Enhanced (27 feat.)	0.953	0.962	+0.009	0.848	0.860

Table 9. TabPFN-v2 vs Random Forest at the same action thresholds (0.285 screening, 0.380 enhanced).

TabPFN beats Random Forest on AUC and F1 for both feature sets, though it trades a small amount of recall (0.84 vs 0.89 at the high-recall threshold) for substantially higher specificity. We keep the Random Forest as the deployed model because its higher recall aligns with the screening-first philosophy and because TabPFN inference is ~430× slower on this hardware (~8.6 s vs ~20 ms for the full 136-patient holdout on CPU). The result is a first-order sanity check that no obvious headroom is being left on the table by the chosen algorithm.

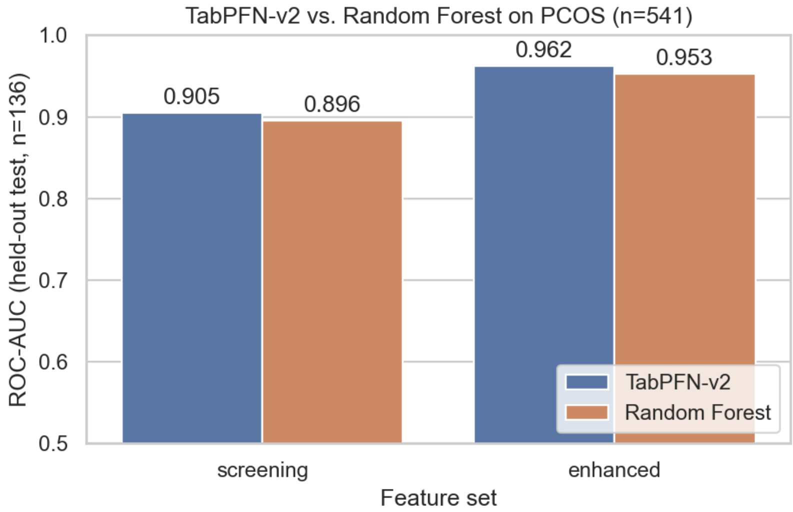


Figure 14. ROC-AUC comparison: TabPFN-v2 (Hollmann et al. *Nature* 2025) vs Random Forest.

10.2 Rotterdam-Criteria Clinical Rule (Teede et al., Fertil Steril 2023)

The international 2023 evidence-based PCOS guideline (Teede et al.) reaffirms the Rotterdam 2003 framework: PCOS-positive if at least 2 of 3 criteria are met — oligo/anovulation, hyperandrogenism, polycystic ovaries. Notebook 13 maps available dataset features to each criterion and benchmarks the hand-coded rule against the enhanced ML model.

Metric	Rotterdam 2-of-3 rule	Enhanced ML model	□ (ML − rule)
Sensitivity	0.795	0.886	+0.091
Specificity	0.913	0.902	–0.011
PPV	0.814	0.812	–0.001
NPV	0.903	0.943	+0.040
F1	0.805	0.848	+0.044
F2	0.799	0.871	+0.072
Accuracy	0.875	0.897	+0.022

Table 10. Hand-coded Rotterdam 2-of-3 rule vs enhanced ML on the same holdout (n=136). The dataset lacks a direct serum-testosterone column, so biochemical hyperandrogenism is approximated using AMH and LH/FSH-related features; this is a documented limitation.

The ML model adds 9.1 percentage points of sensitivity for only a 1.1-point specificity cost — exactly the trade we want from a clinical-decision-support layer that *augments* rather than replaces the guideline-anchored rule.

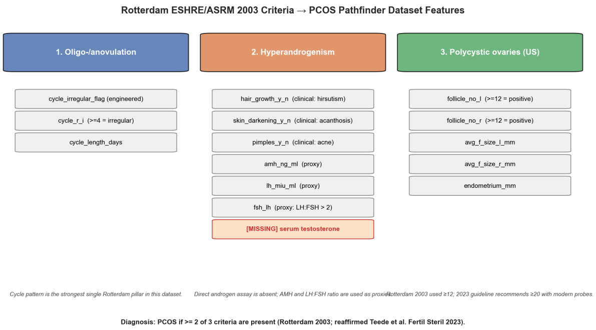


Figure 15. Feature-to-criterion map (3 Rotterdam criteria dataset columns).

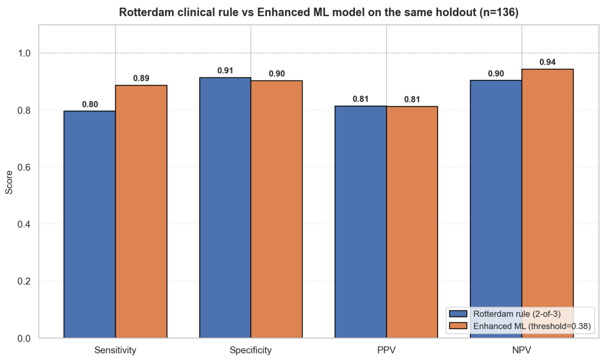


Figure 16. Rotterdam rule vs enhanced ML, side-by-side metrics.

11. Explainability

Explainability is treated as a core product feature, not an afterthought.

11.1 Global Drivers

Global feature importances are exported per model. The screening model leans on symptom presentation (skin darkening, hair growth, weight gain) and basic vitals; the enhanced model is dominated by ultrasound (follicle counts) once those features are available.

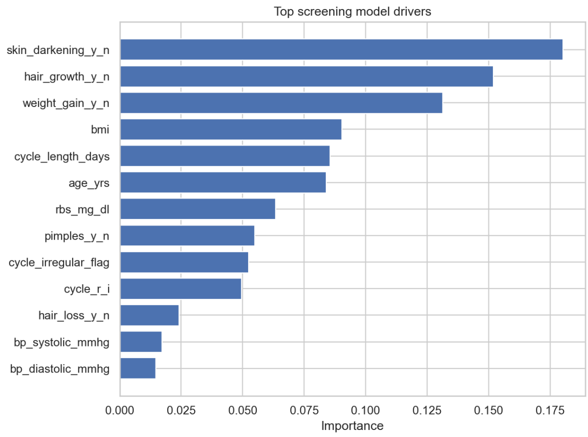


Figure 17. Screening global drivers.

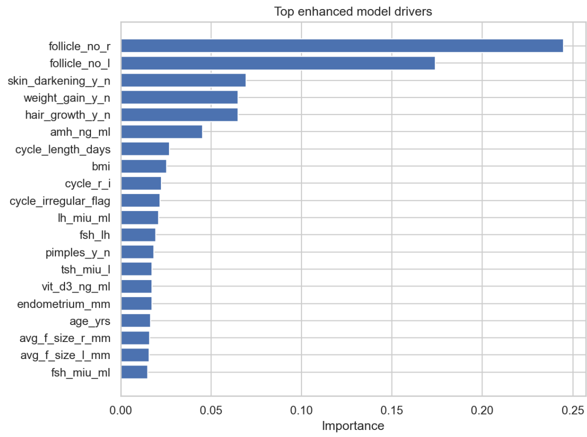
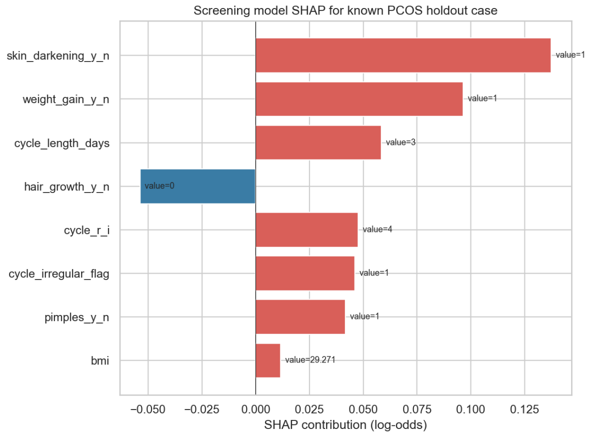


Figure 18. Enhanced global drivers.

11.2 Patient-Level SHAP Attributions

For tree models, `shap.TreeExplainer` produces fast and exact attributions. For logistic-regression fall-backs, each model artifact ships with a 100-row transformed training background that `shap.LinearExplainer` consumes. Demo cases are drawn from the held-out test slice (reproduced from the same random seed), so the explanations describe rows the model never saw during training.



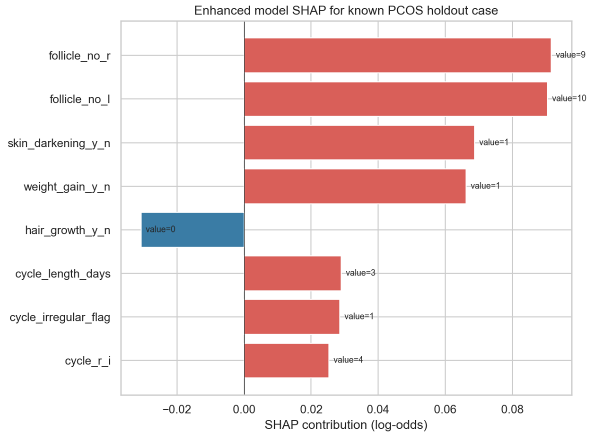


Figure 19. Screening SHAP, holdout PCOS-positive case.Figure 20. Enhanced SHAP, holdout PCOS-positive case.

12. Differential-Diagnosis Prompt

The endometriosis-overlap model is intentionally modest. It is trained on the supplementary synthetic dataset (10,000 rows) using six features (age, BMI, menstrual irregularity, chronic pain level, hormone-level abnormality, infertility) and reaches ROC-AUC 0.66 on a held-out split — a level that reflects the limits of the underlying synthetic data rather than a modelling failure. In the prototype, a positive overlap score raises a warning that the clinician should consider a broader gynaecologic differential workup, especially when chronic pain and infertility are reported alongside menstrual irregularity. It is *not* described as a validated endometriosis diagnostic model.

13. Prototype

The Streamlit prototype at `src/app.py` loads the three joblib model artifacts and presents three tabs:

- **Patient assessment** — intake form with collapsible sections for labs/ultrasound and differential-workup inputs, followed by stacked screening, enhanced, diagnostic-completeness, and endometriosis-overlap results. Each PCOS result shows probability, action threshold, threshold-anchored risk tier, and top SHAP drivers. The checklist adds Rotterdam-style workflow prompts, a missing-test caveat for biochemical hyperandrogenism, a metabolic follow-up prompt, and a lean-PCOS safety warning.
- **Population view** — symptom prevalence by PCOS status (source cohort, n=541) and headline model metrics.
- **About** — plain-language model descriptions, threshold strategy, and caveats.

Risk tiers are anchored to each model's threshold so the displayed tier never contradicts the action flag:

- **Low**: probability below threshold.
- **Moderate**: probability from threshold to threshold + 0.20 (capped at 0.90).
- **High**: probability above the moderate band.

14. Biological Rationale from Single-Cell Feature Lists

Notebook 06 extracts feature symbols from the first available PCOS single-cell sample and checks for the presence of a curated list of literature-supported PCOS genes across steroidogenesis (CYP17A1, CYP19A1, HSD3B2, STAR), androgen signalling (AR, SRD5A1, SRD5A2, HSD17B3), follicular development (FSHR, LHCGR, GATA4), AMH signalling (AMH, AMHR2), insulin signalling (INSR, IRS1, IRS2, INS), and PCOS GWAS loci (DENND1A, THADA, FSHB). All 20 curated genes appear in that sampled feature list, giving the slide deck a defensible biological rationale for the clinical feature families the model uses. No QC, normalisation, integration, clustering, or differential expression analysis is claimed.

15. Limitations and Risk Controls

- **Small, single-source PCOS dataset.** 541 patients from 10 hospitals in Kerala, India is not enough for deployment-level validation, and the geography may not generalise.
- **Synthetic endometriosis data.** The differential module is a workflow prompt, not a diagnostic claim.
- **Equity variables are unavailable.** Race, income, geography, and access-to-care measures are missing, so the fairness audit in §9 uses only age and BMI.
- **Subgroup recall gaps.** A 25-point BMI recall gap in screening (lowest in the normal-BMI band) is an honest finding and a deployment risk; mitigation is subgroup-specific monitoring and external validation on a more balanced cohort.
- **Missing serum testosterone.** The Rotterdam benchmark in §10.2 approximates biochemical hyperandrogenism with AMH and LH/FSH features because direct testosterone is not in the dataset.
- **Clinical framing.** All outputs are framed as risk, triage, and decision support, not diagnosis.
- **Version sensitivity.** scikit-learn model artifacts are sensitive to version skew; `requirements.txt` pins the core artifact-loading stack and constrains TabPFN to the open-weights checkpoint line used for notebook 10.
- **Threshold philosophy.** Tiered thresholds favour sensitivity; clinicians who prefer specificity-first triage can switch to the balanced threshold in Table 4 without retraining.

16. Reproducible Artifacts

Artifact	Purpose
notebooks/01_pcos_data_audit_and_eda.ipynb	Loading, cleaning, audit, EDA, processed CSV export.
notebooks/02_train_pcos_screening_model.ipynb	Training, threshold selection, evaluation: screening.
notebooks/03_train_pcos_enhanced_model.ipynb	Training and evaluation: labs/ultrasound model.
notebooks/04_train_endometriosis_overlap_model.ipynb	Synthetic-data differential prompt model.
notebooks/05_thresholds_explainability_and_demo.ipynb	SHAP, held-out demo cases, model card.
notebooks/06_single_cell_gene_peek.ipynb	Lightweight biological-rationale gene-list check.
notebooks/07_uncertainty_quantification.ipynb	Bootstrap 95% CIs, paired AUC-difference test.
notebooks/08_calibration_analysis.ipynb	Brier, ECE, Platt and isotonic recalibration.
notebooks/09_decision_curve_analysis.ipynb	Net-benefit curves vs treat-all / treat-none baselines.
notebooks/10_tabpfn_benchmark.ipynb	TabPFN-v2 (Nature 2025) vs Random Forest benchmark.
notebooks/11_subgroup_fairness.ipynb	Subgroup recall audit by age and BMI.
notebooks/12_conformal_prediction.ipynb	Split-conformal prediction sets with coverage check.
notebooks/13_rotterdam_alignment.ipynb	Rotterdam 2-of-3 rule benchmark vs enhanced ML.
src/app.py	Streamlit clinician-facing prototype.
outputs/models/*.joblib	Saved model artifacts (with SHAP backgrounds).
outputs/metrics/*	Held-out metrics, threshold trade-offs, SHAP tables, CIs, calibration, DCA, fairness, conformal, Rotterdam, TabPFN.
scripts/create_training_notebooks.mjs	Deterministic notebook regenerator for notebooks 01-06.
report/build_pdf.py	Reportlab generator for this PDF.
PROJECT_PLAN.md, README.md, SUBMISSION.md	Plan, quickstart, packaging checklist.

17. Reproducing This Project From Scratch

This section contains everything a fresh reader needs to rebuild every number, figure, model artifact, and PDF in this report from the source data alone. The pipeline is deterministic given the random seeds documented below.

17.1 Environment

- Python 3.12 (tested) on Windows or POSIX. Node.js ≥ 18 is only needed to regenerate notebooks 01–06 from the JS template.
- Install dependencies: `pip install -r requirements.txt`. The file pins the runtime stack (pandas 2.2.3, numpy 2.2.6, scikit-learn 1.6.1, shap 0.51.0, reportlab 4.4.9, …) and constrains TabPFN with `tabpfn<7` so the open-weights checkpoint is used.
- Random seed: `RANDOM_STATE = 42` is used everywhere — `train_test_split`, `StratifiedKFold`, `cross_val_predict`, model constructors, bootstrap resampling (`np.random.default_rng(seed=42)`), and conformal proper-train/calibration split.

17.2 Source Data and Cleaning

Place either the extracted source files in `_read_extract/` or the original `OneDrive_1_5-11-2026.zip` at the project root. Notebook 01 will auto-extract the two required tabular files if only the zip is present:

- `(Main_Dataset)_PCOS_data_without_infertility.xlsx` — required PCOS clinical dataset (Kerala, n=541).
- `(Supplementary_Dataset)_structured_endometriosis_data.csv` — synthetic endometriosis differential dataset (10,000 rows).
- `(Supplementary_Dataset)_PCOS_single_cell_data.zip` and `(Supplementary_Dataset)_Endometrium_single_cell_data.zip` — optional, only notebook 06 reads them.

Notebook 01 then runs the cleaning recipe documented in §3: column-name standardisation, numeric coercion with logged failures, drop of `sl_no` / `patient_file_no` / `blood_group` as an ordinal numeric input / `marriage_status_yrs`, engineering of `cycle_irregular_flag`, and conservative outlier capping on FSH / LH / FSH÷LH / vitamin D3 / progesterone. The cleaned table is written to `outputs/pcos_cleaned.csv` and is the input to every downstream notebook.

17.3 Pipeline Execution Order

All thirteen notebooks are deterministic and idempotent. Run order matters because notebooks 05, 07–13 all load the joblib artifacts produced by 02–04.

Step	Command	Produces
0. Regenerate notebooks 01–06 (optional, only after editing the .mjs)	node scripts/create_training_notebooks.mjs	notebooks/01–06.ipynb
1. Run notebook 01	python -m nbconvert --to notebook --execute --inplace notebooks/01_pcos_data_audit_and_eda.ipynb	outputs/pcos_cleaned.csv, outputs/metrics/pcos_*.csv, EDA figures
2. Run notebooks 02 and 03	... --inplace notebooks/02_train_pcos_screening_model.ipynb && ... --inplace notebooks/03_train_pcos_enhanced_model.ipynb	outputs/models/pcos_{screening,enhanced}_model.joblib (+ SHAP backgrounds)
3. Run notebook 04	... --inplace notebooks/04_train_endometriosis_overlap_model.ipynb	outputs/models/endometriosis_overlap_model.joblib
4. Run notebook 05	... --inplace notebooks/05_thresholds_explainability_and_demo.ipynb	SHAP per-patient figures, demo case JSON, model card
5. Run notebooks 06–13 in any order	for nb in notebooks/0[6789]*.ipynb notebooks/1*.ipynb; do python -m nbconvert --execute --inplace --to notebook $nb; done	Single-cell genes, bootstrap CIs, calibration, DCA, TabPFN, subgroup, conformal, Rotterdam outputs
6. Build the PDF	python report/build_pdf.py	report/pcos_pathfinder_report.pdf (this file)
7. Launch the prototype	streamlit run src/app.py	Local clinician interface on http://localhost:8501

17.4 Methodology Procedures (concise but complete)

Train/test split

`train_test_split(X, y, test_size=0.25, stratify=y, random_state=42)`. Every downstream notebook reproduces the same split from this seed before loading the saved models.

Threshold selection without test leakage

For each chosen model: compute out-of-fold positive-class probabilities on the training set with `cross_val_predict(method="predict_proba", cv=StratifiedKFold(5, shuffle=True, random_state=42))`. Sweep candidate thresholds in [0.05, 0.95] step 0.005. Pick the largest threshold satisfying training-CV recall ≥ 0.90; if no candidate qualifies, fall back to the highest-F1 threshold. The held-out test set is touched exactly once after refitting on the full training data.

Bootstrap 95% CIs (notebook 07)

Draw 2000 resamples from the holdout with replacement using `np.random.default_rng(seed=42).choice(n_holdout, n_holdout, replace=True)`. For each resample, compute every metric at the artifact-stored threshold. Skip resamples that collapse to a single class (AUC undefined). Report the 2.5th and 97.5th percentile of valid resamples as the 95% CI, plus the bootstrap standard error. The paired AUC-difference test reuses the same 2000 resampled index vectors for both models; report the percentile CI of the difference and the two-sided empirical p-value as `2 × min(P(Δ≤0), P(Δ≥0))`.

Calibration (notebook 08)

Split the 405-row training set into 70% proper-train and 30% calibration with the same random state. Wrap the base classifier in a `FrozenEstimator` (sklearn ≥ 1.6) or use `cv="prefit"` on older sklearn. Fit `CalibratedClassifierCV(method="sigmoid")` for Platt and `method="isotonic"` for isotonic. Compute Brier and ECE (10 equal-width bins) on the held-out test set for raw, Platt-recalibrated, and isotonic-recalibrated probabilities.

Decision Curve Analysis (notebook 09)

For threshold probability p_t across [0.01, 0.99] in 0.005 steps, net benefit is $NB(p_t) = TP/N - FP/N \times (p_t / (1 - p_t))$. The *useful range* is the longest contiguous interval where NB_{model} exceeds both $\max(NB_{treat\text{-}all}, 0)$ and 0.

Conformal prediction (notebook 12)

Split-conformal: from the 405-row training set, hold out 30% (n=122) as a calibration slice and refit a clone of the enhanced pipeline on proper-train only (n=283). On the calibration slice compute nonconformity scores $s = 1 - p_{model}(y\_true \mid x)$. Set q-hat to the quantile of s at level ceil((n_cal + 1) × (1 − □)) / n_cal with □ = 0.10. For each holdout patient the prediction set is the labels {y : 1 − p(y | x) ≤ q-hat}. Coverage validity guarantee is marginal over the holdout.

Subgroup fairness (notebook 11)

Stratify the holdout by age band (`<25, 25–34, 35+`) and BMI category (`underweight, normal, overweight, obese`). For each cell with at least one positive and one negative, draw 1000 stratified bootstrap resamples (preserving the cell's class counts) and report the 95% percentile CI for recall, specificity, PPV, NPV. Cells with n < 10 or zero positives are flagged `insufficient_n`. *Max recall gap* is reported per axis per model.

Rotterdam rule (notebook 13)

Hand-coded mapping to the three Rotterdam criteria using *training-fold* percentiles only (no test leakage on the AMH or LH thresholds): (1) oligo-/anovulation = `cycle_irregular_flag` or `cycle_r_i` ≥ 4; (2) hyperandrogenism = clinical (`hair_growth` | `skin_darkening` | `pimples`) or biochemical proxy (AMH ≥ train P75 or LH ≥ train P75 or FSH/LH < 0.5); (3) polycystic-ovary morphology = `follicle_no_l` ≥ 12 or `follicle_no_r` ≥ 12. Rule predicts PCOS-positive if any 2 of 3 criteria are met. The dataset lacks a direct serum-testosterone column; biochemical hyperandrogenism is therefore an approximation, documented as a limitation in §15.

17.5 Output Files

Each notebook ends with an `assert` cell that confirms its output files exist. The complete inventory:

Path	Contents
outputs/pcos_cleaned.csv	Processed PCOS table after cleaning (input to all downstream models).
outputs/models/pcos_screening_model.joblib	Pipeline (median imputer + classifier), feature list, threshold, metrics, 100-row SHAP background.
outputs/models/pcos_enhanced_model.joblib	Same artifact shape, with labs+ultrasound feature set.
outputs/models/endometriosis_overlap_model.joblib	Differential-prompt model on synthetic data.
outputs/metrics/{screening,enhanced,endometriosis}_holdout_metrics.json	Point estimates and confusion matrices at the action threshold.
outputs/metrics/holdout_bootstrap_cis.{json,csv}	2000-resample 95% CIs and paired AUC-difference test.
outputs/metrics/calibration.json	Brier and ECE for raw / Platt / isotonic, per model.
outputs/metrics/dca.json	Net-benefit grid, treat-all curve, useful range, per model.
outputs/metrics/tabpfn_comparison.json	TabPFN-v2 vs Random Forest head-to-head; status flag for skipped runs.
outputs/metrics/subgroup_performance.{json,csv}	Per-subgroup metrics with bootstrap CIs and max recall gap.
outputs/metrics/conformal_coverage.json	Target/empirical coverage, q-hat, set-size distribution, worked examples.
outputs/metrics/rotterdam_comparison.json	Rotterdam-rule metrics vs enhanced ML, with explicit rule definition and feature mapping.
outputs/metrics/pcos_model_card.md	Concise model card emitted by notebook 05 (intended use, data, limitations).
outputs/figures/*.png	All plots referenced from this report and the deck.

18. Conclusion

PCOS Pathfinder is not an automated diagnosis. It is a practical, explainable, tiered workflow that (a) identifies high-risk patients earlier with recall ≥ 0.88 and NPV ≥ 0.93; (b) demonstrates a AUC improvement of the enhanced over the screening model on this split (□=+0.057, p≈0.015); (c) ships calibrated probabilities, conformal coverage guarantees, decision-curve utility, subgroup fairness numbers, a recent-SOTA TabPFN benchmark, and a Rotterdam clinical-rule benchmark; (d) flags its own equity limitations honestly. The cleaning rules, threshold strategy, and SHAP attributions are reproducible end-to-end from the thirteen numbered notebooks; the Streamlit prototype loads the same artifacts a clinician would use; and the limitations are stated up-front in the model card.

19. Selected References

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